

Homework 1 — Image Processing

1. Environment

```
python >= 3.9
numpy
opencv-python
```

```
pip install -r requirements.txt
```

Every script runs from the package root and reads from `data/`, writing to `out/`:

```
python p1_smoothing.py
python p2_upsampling.py
python p3_downsampling.py
python p4_pyramid.py
python p5_edges.py
python check.py           # self-check, see section 4
```

The reference implementation takes well under a minute for all five scripts on a laptop. `p1_smoothing.py` and `p3_downsampling.py` are the slow ones, because they filter a 1920x1080 image with large kernels.

2. What you implement

File	Functions	Points
<code>p1_smoothing.py</code>	<code>create_gaussian_kernel</code> , <code>pad</code> , <code>filtering</code>	25
<code>p2_upsampling.py</code>	<code>upsample_nearest</code> , <code>upsample_bilinear</code>	20
<code>p3_downsampling.py</code>	<code>downsample_naive</code> , <code>downsample_prefiltered</code>	15
<code>p4_pyramid.py</code>	<code>build_gaussian_pyramid</code>	15
<code>p5_edges.py</code>	<code>sobel_kernels</code> , <code>compute_gradient</code> , <code>create_dog_kernels</code> , <code>non_maximum_suppression</code> , <code>hysteresis_threshold</code>	25

Each function body is marked with `TODO: implement this` followed by `raise NotImplementedError`. Delete both lines and write your code there. Do not change the function signatures, the surrounding code, or `imutils.py`; the self-check and the grading assume them.

Read the questions in `report_template.docx` before you start. They are where most of the learning is; the code exists to produce the evidence for them.

3. Rules

You may use

- `numpy` for everything,
- `cv2.imread` and `cv2.imwrite` (already called for you in `imutils.py`),

- `matplotlib`, if you prefer it for the figures in the report.

You may not use, anywhere in `p1` – `p5` :

`cv2.filter2D`, `cv2.GaussianBlur`, `cv2.blur`, `cv2.sepFilter2D`, `cv2.getGaussianKernel`,
`cv2.resize`, `cv2.pyrDown`, `cv2.pyrUp`, `cv2.Sobel`, `cv2.Scharr`, `cv2.Laplacian`, `cv2.Canny`,
`numpy.pad`, `numpy.convolve`, `scipy.ndimage`, `scipy.signal`, `skimage`.

These are exactly the functions the homework asks you to write. They appear in `check.py`, which is the one file allowed to call them.

4. Self-check

```
python check.py
```

The script compares your output with the OpenCV built-ins on `data/cat.png` and prints one line per check. It runs in a couple of seconds and reports `SKIP` for anything you have not implemented yet, so it is useful from the first function onwards.

`check.py` is a debugging aid, **not the grader**.

- Passing every check does not mean full marks: the figures and the written answers count for as much as the code.
- Failing a check does not mean zero. If you made a different but defensible choice, say so in the report and you keep most of the credit.

Paste the full output into section 2 of your report.

5. Conventions you must follow

These are fixed so that every submission can be compared against the same reference. Getting them wrong changes every number in your report.

Resampling coordinates. For an output index `d` on one axis, with `scale = old_size / new_size`, the source coordinate is

```
u = (d + 0.5) * scale - 0.5
```

clamped to `[0, old_size - 1]`. This aligns the centres of the two grids (Lecture 2, slides 89–93). `u = d * scale` shifts the image by half a pixel.

Padding. Two modes, as in Lecture 2 (slides 17–18):

```
zero      0 0 0 | a b c d | 0 0 0
replicate a a a | a b c d | d d d
```

Filtering is correlation, not convolution: the kernel is not flipped.

Kernels. A smoothing kernel is normalised to sum to 1. A derivative kernel sums to 0 and is not normalised.

Data types. Compute in `float64`. `imutils.imwrite` clips to `[0, 255]` and rounds when it writes a PNG; do not clip earlier.

6. Submission

Submit a single archive named `HW1_<student id>_<name>.zip` containing:

```
HW1_<student id>_<name>/
├─ p1_smoothing.py ... p5_edges.py    your implementation
├─ out/                               the images your scripts produced
└─ report.pdf                         at most 8 pages
```